Problem Sheet: Week 7

1. Let $\underline{X}$ be a random sample from a Poi(λ) distribution. Show carefully that the conjugate prior for λ is the Gamma(α, β) distribution.

   For a random sample of size 10, $\sum x_i = 28$, being relatively ignorant about λ you choose your prior to be $\lambda \sim \text{Gamma}(0.5, 0.5)$.

   Find the posterior distribution for λ.

2. Suppose that in the situation of Q1 instead of observing your random sample of size 10 at a single time you received the data in two steps.

   First you observe $\sum_1^5 x_i = 12$ and then, at a later time, $\sum_6^{10} x_i = 16$.

   Using the same prior as in Q1:

   (i) Find the posterior density $\pi(\lambda | \sum_1^5 x_i = 12)$.

   (ii) Use this posterior as a prior distribution for a Bayesian analysis of the second observation $\sum_6^{10} x_i = 16$.

   (iii) Verify that the posterior distribution after the second analysis is the same as that obtained in Q1 where all observations are analysed in a single step.

3. Let $\underline{X} = (X_1, X_2, \ldots, X_n)$ be a random sample from the Pareto distribution with p.d.f.

$$f(x) = \theta(1+x)^{-(\theta+1)}, \qquad 0 < x < \infty.$$

   Show that the Gamma distribution is the conjugate prior for this distribution by proving that if the prior distribution of θ is Gamma(α, β), then the posterior is

$$\text{Gamma}(\alpha + n, \beta + t(\underline{X})),$$

   where $t(\underline{X}) = \sum_i \log(1 + X_i)$.

4. Let $\underline{X}$ denote a random sample from the U(0, θ) distribution. Show that the conjugate prior for this case is from the Pareto family with p.d.f.

$$\pi(\theta; \alpha, \beta) = \frac{\beta \alpha^\beta}{\theta^{\beta+1}}, \qquad \theta > \alpha.$$

5. Let $y = (x_1, \ldots, x_n)$ be a random sample from a N(μ, σ^2) distribution where σ^2 is known. Suppose we use a N(α, β) prior for μ. Use Bayes' theorem to derive the posterior distribution $\pi(\mu|y)$ and comment on the form of the posterior expectation of μ.